

Course Outline

1. What is AI?
 - a. Definitions and views
 - b. Machine learning
 - c. Reinforcement learning
 - d. Foundation models (a teaser — developed fully in Section 2)
 - e. Key facts and trends
 - f. Economics in AI
 - g. Economics of AI
2. Foundation models
 - a. What is a foundation model? Capabilities
 - b. Competition in the vertical stack
 - c. What's new: industry features (proliferation, entry barriers, cost-per-performance, converging performance, winner-take-all vs. commoditization)
 - d. AI partnerships
 - e. Partnerships: theory (foreclosure, technology and talent pooling)
3. AI, jobs, and growth
 - a. AI as a general-purpose technology
 - b. Jobs and tasks before LLMs: exposure and evidence
 - c. Jobs after LLMs: exposure
 - d. Systematic evidence (RCTs: customer support, software development)
 - e. Macroeconomics: task-based model and labor demand
 - f. Macroeconomics: productivity and growth
 - g. Adoption
4. Reinforcement learning
 - a. The Markov decision process framework
 - b. Policies, values, and the Bellman equation
 - c. Model-free methods and Q-learning
5. Agentic markets: an overview
6. Agentic sellers: pricing algorithms
 - a. Introduction: algorithms in markets
 - b. Pricing theory: commitment and competition
 - c. Algorithmic collusion (Calvano, Calzolari, Denicolò, Pastorello)

- d. Empirical evidence (German retail gasoline)
 - e. Field experiment on Amazon
- 7. Agentic platforms: recommender systems
 - a. Introduction and sources of power
 - b. Quantifying the power of Spotify
 - c. Recommender systems: definition and economics
 - d. Recommender systems and competition
- 8. Agentic platforms: choice manipulation
 - a. Framework: manipulating choices
 - b. Inflated recommendations (Peitz and Sobolev)
 - c. Self-preferencing (Aridor and Gonçalves)
 - d. Playing it safe (Calvano and Jullien)
- 9. Agentic buyers
- 10. Algorithms and public policy
 - a. Broad policy challenges
 - b. Why is AI different?
 - c. The policy landscape: DMA, DSA, AI Act
 - d. Application: tackling algorithmic bias
 - e. Application: tackling algorithmic collusion
 - f. Data-driven incumbency advantage
 - g. Looking ahead

About the Instructor

Emilio Calvano (PhD in Economics, Toulouse School of Economics) is Full Professor of Economics at LUISS Guido Carli University in Rome, associate faculty at the Toulouse School of Economics and research fellow of the Centre for Economic Policy Research in London (CEPR) and Einaudi Institute for Economics and Finance in Rome. He is an applied theoretical economist whose research mainly focuses on the theory of Industrial Organization. His research interests include the economics of artificial intelligence, the economics of platforms, information economics and competition policy. His recent work studies the impact of AI powered algorithms (such as pricing software and recommender systems) on digital markets. He has published in top international peer-reviewed outlets such as Science Magazine, American Economic Review, Management Science, American Economic Journal: Microeconomics, Economic Journal and International Journal of Industrial Organization and is associate editor at the Journal of Industrial Economics. In 2023 he has been awarded a 5 year multi-million European Research Council Advanced grant to conduct academic research Artificial Intelligence and Competition.

References and useful study material (* = seen / discussed in class)

On AI and on the Economics of AI

- *Agrawal, A., Gans, J., & Goldfarb, A. (2019). *Economic Policy for Artificial Intelligence*. In *Innovation Policy and the Economy*. University of Chicago Press.
- *Agrawal, A., Gans, J., & Goldfarb, A. (2019b). *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press.
- Agrawal, A., Gans, J., & Goldfarb, A. (2018). *Prediction Machines: The Simple Economics of Artificial Intelligence*. Harvard Business Press.
- *Stanford HAI (2025). *Artificial Intelligence Index Report 2025*. Stanford HAI. (Also: Perrault, R., & Clark, J. (2024). *AI Index Report 2024*.)

Economics in AI

- Hardt, M., Megiddo, N., Papadimitriou, C., & Wootters, M. (2015). *Strategic Classification*. In *arXiv [cs.LG]*. arXiv. <http://arxiv.org/abs/1506.06980>

Foundation Models

- *Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J. Q., Demszky, D., ... Liang, P. (2021). *On the Opportunities and Risks of Foundation Models*. In *arXiv [cs.LG]*. arXiv. <http://arxiv.org/abs/2108.07258>
- *Korinek, A. and Vipra, J. (2024). *Market Concentration Implications of Foundation Models: The Invisible Hand of ChatGPT*. Unpublished manuscript.
- *Stanford HAI (2025). *Artificial Intelligence Index Report 2025*. Stanford HAI. (Source of the industry charts on parameters, training cost, cost-per-performance, and arena convergence.)
- *Rey, P., & Tirole, J. (2007). *A Primer on Foreclosure*. In *Handbook of Industrial Organization* (Vol. 3, pp. 2145–2220). North-Holland. (Vertical-stack foreclosure / exclusive dealing.)
- *Jeon, D.-S., & Lefouili, Y. (2018). [Technology and talent pooling; patent pools.] *RAND Journal of Economics*.

AI, Jobs, and Growth

- *Brynjolfsson, E., Mitchell, T., & Rock, D. (2018). *What Can Machines Learn, and What Does It Mean for Occupations and the Economy?* *AEA Papers and Proceedings*, 108, 43–47.
- *Brynjolfsson, E., Rock, D., & Syverson, C. (2019). *Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics*. In *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press. (The Solow paradox and the “productivity J-curve”.)
- *Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2023). *GPTs are GPTs: An Early Look at the Labor Market Impact Potential of Large Language Models*. In *arXiv [econ.GN]*. arXiv. <http://arxiv.org/abs/2303.10130>
- *Brynjolfsson, E., Li, D., & Raymond, L. (2025). *Generative AI at Work*. *Quarterly Journal of Economics* (R&R). NBER Working Paper 31161. (Customer-support field experiment.)

- *Cui, K. Z., et al. (2025). The Effects of Generative AI on High-Skilled Work: Evidence from Field Experiments with Software Developers. SSRN working paper.
- *Acemoglu, D., & Restrepo, P. (2019). Automation and New Tasks: How Technology Displaces and Reinstates Labor. *Journal of Economic Perspectives*, 33(2), 3–30. (Task-based model of labor demand.)
- *Acemoglu, D. (2024). The Simple Macroeconomics of AI. *Economic Policy*. NBER Working Paper 32487.
- *McElheran, K., Li, J. F., Brynjolfsson, E., Kroff, Z., Dinlersoz, E., Foster, L., & Zolas, N. (2024). AI Adoption in America: Who, What, and Where. *Journal of Economics & Management Strategy*, 33(2), 375–415.

Recommender Systems

- *Aggarwal, C. C. (2016). Recommender Systems: The Textbook. Springer, Cham. <https://doi.org/10.1007/978-3-319-29659-3> (sections of chapters 2, 3).
- *Aguiar, L., & Waldfogel, J. (2021). Platforms, Power, and Promotion: Evidence from Spotify Playlists. *Journal of Industrial Economics*, 69(3), 653–691. (Quantifying the power of Spotify.)
- *Calvano, E., Calzolari, G., Denicolò, V., & Pastorello, S. (2024). Artificial Intelligence, Algorithmic Recommendations and Competition. <https://doi.org/10.2139/ssrn.4448010>
- *Claussen, J., Peukert, C., & Sen, A. (2021). The Editor and the Algorithm: Returns to Data and Externalities in Online News. <https://doi.org/10.2139/ssrn.3479854>
- Fleder, D., & Hosanagar, K. (2009). Blockbuster culture’s next rise or fall: The impact of recommender systems on sales diversity. *Management Science*. <https://pubsonline.informs.org/doi/abs/10.1287/mnsc.1080.0974>
- Lee, D., & Hosanagar, K. (2019). How do recommender systems affect sales diversity? A cross-category investigation via randomized field experiment. *Information Systems Research*. <https://pubsonline.informs.org/doi/abs/10.1287/isre.2018.0800>
- Lee, K. H., & Musolff, L. (2024). *Entry into two-sided markets shaped by platform-guided search*. R&R *Econometrica*.
- Yeomans, M., Shah, A., Mullainathan, S., & Kleinberg, J. (2019). Making sense of recommendations. *Journal of Behavioral Decision Making*, 32(4), 403–414. <https://doi.org/10.1002/bdm.2118>
- Wan, X. (shawn), Kumar, A., & Li, X. (2020). How Beneficial are Recommendations to Consumers? Estimates of Relative Benefits of Product Recommendations to Consumers and Retailers. <https://doi.org/10.2139/ssrn.3702762>

Choice Manipulation

- *Kamenica, E., & Gentzkow, M. (2011). Bayesian Persuasion. *The American Economic Review*, 101(6), 2590–2615. <https://doi.org/10.1257/aer.101.6.2590>
- *Peitz, M., & Sobolev, A. (2022). Inflated Recommendations. Working paper. (Binary recommendation policies; when biased recommendations are profit-maximizing.)
- *Aridor, G., & Gonçalves, D. (2022). Self-Preferencing by a Vertically-Integrated Intermediary. Working paper.
- *Calvano, E., & Jullien, B. (2022). Playing it Safe: Reputation and Cautious Recommendations. Working paper.

Farronato, C., Fradkin, A., & MacKay, A. (2024). Self-Preferencing at Amazon: Evidence from Search Rankings (No. 30894). National Bureau of Economic Research. <https://doi.org/10.3386/w30894>

Reinforcement Learning

*Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press. http://www.academia.edu/download/38529120/9780262257053_index.pdf (parts of chapter 1 and 6)

Application: Pricing Algorithms

*Assad, S., Clark, R., Ershov, D., & Xu, L. (2020). Algorithmic Pricing and Competition: Empirical Evidence from the German Retail Gasoline Market. <https://papers.ssrn.com/abstract=3682021>

*Brown, Z., & MacKay, A. (2021). Competition in Pricing Algorithms, SSRN working paper. <https://doi.org/10.2139/ssrn.3485024>

*Calvano, E., Calzolari, G., Denicolò, V., & Pastorello, S. (2020). Artificial Intelligence, Algorithmic Pricing, and Collusion. *The American Economic Review*, 110(10), 3267–3297. <https://doi.org/10.1257/aer.20190623>

Calvano, E., Calzolari, G., Denicolò, V., & Pastorello, S. (2021). Algorithmic collusion with imperfect monitoring. *International Journal of Industrial Organization*, 102712. <https://doi.org/10.1016/j.ijindorg.2021.102712>

*Chen, L., Mislove, A., & Wilson, C. (2016). An Empirical Analysis of Algorithmic Pricing on Amazon Marketplace. *Proceedings of the 25th International Conference on World Wide Web*, 1339–1349. <https://doi.org/10.1145/2872427.2883089>

Musolff, L. (2022). Algorithmic Pricing Facilitates Tacit Collusion: Evidence from E-Commerce. *Proceedings of the 23rd ACM Conference on Economics and Computation*, 32–33. <https://doi.org/10.1145/3490486.3538239>

*O'Connor, J., & Wilson, N. E. (2021). Reduced demand uncertainty and the sustainability of collusion: How AI could affect competition. *Information Economics and Policy*, 54, 100882. <https://doi.org/10.1016/j.infoecopol.2020.100882>

Banchio, M., & Mantegazza, G. (2024). Artificial Intelligence and Spontaneous Collusion. *R&R American Economic Review*.

*Johnson, J. P., Rhodes, A., & Wildenbeest, M. (2023). Platform design when sellers use pricing algorithms. *Econometrica: Journal of the Econometric Society*, 91(5), 1841–1879. <https://doi.org/10.3982/ecta19978> (Buy box / price-directed prominence.)

Additional useful references

Machine learning and Economics

Athey, S. (2019). 21. The Impact of Machine Learning on Economics. In *The economics of artificial intelligence* (pp. 507–552). University of Chicago Press. <https://www.degruyter.com/document/doi/10.7208/9780226613475-023/html>

Mullainathan, S., & Spiess, J. (2017). Machine Learning: An Applied Econometric Approach. *The Journal of Economic Perspectives: A Journal of the American Economic Association*, 31(2), 87–106. <https://doi.org/10.1257/jep.31.2.87>

Application: Auctions and algorithmic bidding

F. Decarolis and G. Rovigatti “From Mad Men to Maths Men: Concentration and Buyer Power in Online Advertising,” CEPR Discussion Paper 13897, 2019, American Economic Review, forthcoming.

F. Decarolis, Rovigatti, G., Rovigatti, M., & Shakhgildyan, K. (2023). Artificial Intelligence & Data Obfuscation: Algorithmic Competition in Digital Ad Auctions (No. 18009). CEPR Discussion Papers.

B. Edelman, M. Ostrovsky and M. Schwarz "Internet Advertising and the Generalized Second-Price Auction: Selling Billions of Dollars Worth of Keywords" American Economic Review 97 1 242-259 2007. <https://doi.org/10.3386/w11765>

Algorithmic Bias and fairness

*Lambrecht, A., & Tucker, C. (2019). Algorithmic Bias? An Empirical Study of Apparent Gender-Based Discrimination in the Display of STEM Career Ads. *Management Science*, 65(7), 2966–2981. <https://doi.org/10.1287/mnsc.2018.3093>

*Cowgill, B., & Tucker, C. E. (2020). Algorithmic Fairness and Economics. In preparation for the *Journal of Economic Perspectives*. <https://doi.org/10.2139/ssrn.3361280>

Data and incumbency

*Hagiu, A., & Wright, J. (2023). Data-enabled learning, network effects, and competitive advantage. *RAND Journal of Economics*, 54(4), 638–667.

Bandits

Sutton, R. S., & Barto, A. G. (2018). Reinforcement learning: An introduction. MIT press. http://www.academia.edu/download/38529120/9780262257053_index.pdf (parts of chapter 2)

Aridor, G., Mansour, Y., Slivkins, A., & Wu, Z. S. (2020). Competing Bandits: The Perils of Exploration Under Competition. In arXiv [cs.GT]. arXiv. <http://arxiv.org/abs/2007.10144>

Policy

Kleinberg, J., Ludwig, J., Mullainathan, S., & Obermeyer, Z. (2015). Prediction Policy Problems. *The American Economic Review*, 105(5), 491–495. <https://doi.org/10.1257/aer.p20151023>

*Rambachan, A., Kleinberg, J., Mullainathan, S., & Ludwig, J. (2023). An Economic Approach to Regulating Algorithms. <https://papers.ssrn.com/abstract=3597843>